How it fits together

Threat intelligence → detections for Defender XDR, Google SecOps, Splunk and Elastic

AUTOMATED MULTI-VENDOR THREAT HUNTING & DETECTION SUITE

open source · self-hosted · works without AI by default · @cmdaltr

core flow
optional AI (off unless enabled)
config, context and influence
coming soon
needs user interaction

INTERFACES

Web UI FastAPI, server-rendered

archer-cli start / stop / restart supervisor

Console API

MCP server

INPUTS — EVERY RUN

Company profile
Domain → sector, geography, observed tech stack

recommends groups

Hunt
Threat group or alias · CVE(s) · ATT&CK technique IDs

Analyse
Report (PDF/DOCX) or blog URL → techniques and IOCs

optional

Research
Threat Feed: the week's research from ~26 vendor and government sources, tagged by industry, nation and motive. Watchlists flag new posts (optional email).
▸ Your stack filter

GENERATION PIPELINE

01 Procedure retrieval
Group → techniques → procedures

02 Extraction, hygiene + stack filter
Placeholders, generic words and known-good removed; skips what you don't run

06 IOC sweeps branch
Report and CVE indicators, with expiry

IOCs

OPTIONAL AI · off unless enabled
Ask Archer · query-repair and report-triage agents · LLM via local Ollama or a cloud provider

03 Query builders
- Defender KQL
- SecOps YARA-L
- Splunk
- Elastic

04 Validate + save
Conditions that parse but can't fire are removed; the rest join the shared YAML library

05 Correlation engine
Chains detections across tactics on one host or account in a time window

tech stack

feeds the pipeline · checks YAML indicators

context

OUTPUTS

Results
Threat briefing (coverage and gaps) · detections per SIEM · correlations · attack flow (drag-and-drop, STIX export) · IOC flow · validation checklist

Library
Saved detections and correlations

Coverage
ATT&CK matrix

Flows
Detectable vs not, per attack flow

CONFIG — ONE-TIME SETUP

Profile
SIEM platforms · Threat Feed industries · API keys

bespoke/ mappings
Org-specific index names and overrides, kept out of git

Technology stack
Platforms and products you run

EMULATE

Which logs record each technique (native OS logs first, Sysmon and EDR optional), by technique, group or data source · indicators checked against the recordings · Defender XDR's built-in alerting per technique

TOOLS

Regex tester RE2 and PCRE2 run in their real engines

MITRESaw ATT&CK procedures and their indicators

Cyber Powers map National cyber capability on a world map

BACKEND · INTELLIGENCE SOURCES context, not inputs

storage: SQLite

ATT&CK + MITRESaw
STIX bundle · MITRESaw (git submodule): procedure extraction across direct, campaign and software attribution, with an evidence workbook

NVD + public PoCs
CVE research and exploit indicators

OSINT rule index
Elastic, Splunk, Sentinel and SecOps public rules, plus their documented false positives

LOLBAS
LOLDrivers
LOLRMM

Known-good lists
First-party domains · DNS resolvers · empty-file hashes

Recorded attack telemetry
Splunk attack_data · OTRF Security-Datasets · EVTX-to-MITRE-Attack · macOS-ATTACK-DATASET

Microsoft GUIDE
Defender XDR's own alerting per technique across 5,000+ organisations, with triage grades

Your lab
COMING SOON
Atomic Red Team runs on your test VMs, with Defender for Endpoint's rows

Test VMs
COMING SOON
Windows · Linux · macOS → record-atomics

archer-cli — the same pipeline from a terminal

one command: archer-cli · console modules · python -m maintenance jobs

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core flow
optional AI (off unless enabled)
config, context and influence
coming soon
dimmed = web UI only
needs user interaction

INTERFACES

archer-cli start|stop|restart|run|status|logs · serve · clear-cache · install · docker
--port --host --reload/--no-reload --ollama/--no-ollama --ollama-models DIR --skip-index
logs [archer|ollama|supervisor] -f -n

archer-cli console
use <module> · set/unset <OPTION> · show options|modules
info · search · run · back
ARCHER_URL · ARCHER_TOKEN (or ARCHER_EMAIL / ARCHER_PASSWORD)

archer-cli preview <group>
--siem mde,secops,splunk,elastic --index/-i NAME
--limit N --output/-o FILE

archer-cli mcp
MCP server: hunt_* · wait_for_run · get_run_results
save_query_fix · prompts repair/ask/triage

archer-cli offline generate|correlate|hunt (emulate: coming soon) — the pipeline without the server · generate --correlate --llm/--no-llm --hypothesis · correlate --min-queries --max-queries --yaml-dir --output

INPUTS — EVERY RUN

Company profile
use intel/company DOMAIN
INDUSTRY GEOGRAPHY TECH_STACK
SUPPLY_CHAIN GROUPS (top N|names|0)

GROUPS

Hunt
use hunt/group GROUP · hunt/cve CVE
use hunt/technique TECHNIQUES
EXCLUDE (secops,mde,splunk,elastic)
MERGE_BY_TACTIC MERGE_BY_PARENT BRIEFING

Analyse
use intel/report URL | FILE
MODE all|iocs|review TECHNIQUES USE_AI

optional

Research
use intel/feed SEARCH SHOW relevant|all
INDUSTRIES REFRESH WAIT

use research/actors KEYWORDS TOP
MODE AND|OR SCOPE ACTORS EXPORT xlsx

GENERATION PIPELINE

01 Procedure retrieval
GROUP · HYPOTHESIS

02 Extraction, hygiene + stack filter
LOL default on · OSINT
OSINT_COMPARE
LOL = LOLBAS/LOLDrivers/LOLRMM
stack: from profile (no option)

06 IOC sweeps branch
intel/report MODE iocs · hunt/cve

IOCs

03 Query builders
SIEM / EXCLUDE · INDEX

04 Validate + save
VALIDATE Docker SIEMs
python -m archer.contract validate|lint|report

DRY_RUN stdout only
writes data/groups/*.yaml

05 Correlation engine
CORRELATE MIN_RELEVANCE
MIN_QUERIES MAX_QUERIES

OPTIONAL AI · off unless enabled
MODE llm|fallback|standard · PROVIDER assigned|none|anthropic|openai|google|deepseek|qwen|grok
MODEL · MAX_CALLS
use ai/ask · ai/briefing GROUP TOP_N · agent/repair RUN SCOPE LIMIT
agent/ask · agent/triage URL · agent/history SESSION CONTINUE APPROVE ask|all

WAIT · OUTPUT <zip> — every server-driven module

tech stack

feeds the pipeline · checks YAML indicators

context

OUTPUTS

Results
use results RUN LATEST QUERIES OUTPUT <zip>

Library WEB UI ONLY
Saved detections and correlations

Coverage WEB UI ONLY
ATT&CK matrix

Flows WEB UI ONLY
Detectable vs not, per attack flow

CONFIG — ONE-TIME SETUP

Profile
use ai/settings ASK_PROVIDER ASK_MODEL
DETECTION_PROVIDER DETECTION_MODEL
MAX_CALLS LOCAL_ENDPOINT LOCAL_MODEL
REMEMBER

bespoke/ mappings
files only · gitignored
bespoke/mappings · prompts · query_templates

Technology stack
use intel/company TECH_STACK
otherwise web UI only

EMULATE

python -m archer.telemetry --build
[--source splunk|otrf|evtx|macos|lab]
python -m archer.telemetry --guide <zip|folder|csv>
--mde-export <run dir>

TOOLS

Regex tester WEB UI ONLY

MITRESaw python -m archer.mitre.procedures [--full] [-o FILE]
incremental by default

Cyber Powers map WEB UI ONLY

BACKEND · INTELLIGENCE SOURCES context, not inputs

storage: SQLite

ATT&CK + MITRESaw
STIX bundle · MITRESaw (git submodule): procedure extraction, evidence workbook
python -m archer.mitre.procedures

NVD + public PoCs
CVE research and exploit indicators
fetched on demand

OSINT rule index
Elastic, Splunk, Sentinel, SecOps public rules + documented false positives
scripts/data/update_osint.py
--force --no-index --elastic --splunk
--azure-sentinel --chronicle

LOLBAS LOLDrivers LOLRMM
fetched by update_osint.py

Known-good lists
First-party domains · DNS resolvers · empty-file hashes
fetched on demand

Recorded attack telemetry
Splunk attack_data · OTRF Security-Datasets · EVTX-to-MITRE-Attack · macOS-ATTACK-DATASET
python -m archer.telemetry --build

Microsoft GUIDE
Defender XDR's own alerting per technique, 5,000+ organisations
python -m archer.telemetry --guide

Technique yield
python -m archer.build.technique_yield
--rebuild · --reset-llm [TID…]

Your lab
COMING SOON
Atomic Red Team runs on your test VMs, with Defender for Endpoint's rows

Test VMs
COMING SOON
Windows · Linux · macOS
→ record-atomics